

Yash Mahadev Parab

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Engineering Management Graduate (RWTH Aachen) with core competencies in market research, strategy development, project management & entrepreneurial finance. Leveraging business acumen with data analysis capabilities to build and deploy digital management solutions.

Key Skills

Business Development | Market Analysis | Strategy Development | Project Management | Data Analysis | Stakeholder Collaboration | KPI Monitoring & Reporting | Process Optimization | Entrepreneurial Finance

Professional Experience

Working Student – Business and Market Research Analyst Feb 2025 – Present
Fraunhofer FHR (Industrielle Hochfrequenzsysteme [IHS] Operations) Wachtberg, Germany

- Delivered end-to-end 20+ market intelligence reports on high-tech sectors (New Space Tech, Military Drones, Electronic Warfare, Urban Mobility, AI-Intralogistics) to synthesize insights into actionable strategy recommendations.
- Produced strategic reports and executive presentations for 100+ competitors and product profiling projects utilizing premier industry sources (Yole, Frost&Sullivan) to bolster business development.
- Uncovered technology trends, innovation gaps and key players from 100M+ patents and publications by using Python and R based data analysis.
- Cut manual scouting time by ~90% and identified ~700 target opportunities for the team's marketing and networking plan, by automating data monitoring via a custom Python web scraper across ~3K events.

Research Assistant – Sustainability Analyst and Business Development Aug 2023 – Dec 2023
Production Engineering of E-Mobility Components (PEM RWTH) Aachen, Germany

- Built and presented interactive Power BI dashboards to stakeholders, validating circular-economy business models for 2nd-life batteries.
- Identified 8-9% potential energy savings across 12+ e-mobility OEMs by designing and performing competitor analysis and financial assessment framework.
- Delivered audit-ready ESG reporting and strategy by owning the emissions data evaluation process through the full EV battery value chain.

Project Engineer – Process Automation and Optimization Jun 2020 – Sep 2021
R.K. Control Instruments Pvt. Ltd. Thane, India

- Improved cross-functional collaboration and process efficiency by designing standardized analytical frameworks adopted team-wide.
- Led a key automation and optimization project by building an Excel-Python solution for 500+ data files, that cut cycle time from 15 mins to 4 mins (~75%), directly enhancing business decisions and operational cost management.

Master's Thesis

Typologies of Business Angels: Classification Criteria and Differences in Investment Behavior

Live App: [BA3F \(Business Angels Funding Fit Finder\)](#) (Note: app sleeps when idle, kindly wait ~10 seconds)

- Developed a classification framework for the informal venture capital market through structured analysis across 5 databases.
- Identified 5 typological dimensions – demographics, experience, psychology, styles, organization) by mapping behavioral differences in risk tolerance, syndication and investment preferences.
- Operationalized framework into a live LLM-powered classifier (BA3F) that reads a public investor bio and returns a typology match plus pitch-tailoring guidance, reducing founder guesswork on who, how, and what to pitch.
- Addressed information asymmetry and angel-market heterogeneity for entrepreneurs and policymakers by delivering evidence-grounded outputs translated from deterministic findings in the exhaustive literature review.

Projects

Earnings Reaction Predictor (Personal): Equity Earnings-Surprise Analytics

Live App: [ERP \(Earnings Reaction Predictor\)](#) (Note: app sleeps when idle, kindly wait ~10 seconds)

- Built and deployed a dual-model ML pipeline predicting the direction and magnitude of post-earnings stock moves across 403 earnings events, 51 US companies and 10 sectors.
- Validated on 100% out-of-time 2026 data (pre-2026 train, unseen 2026 test), delivering an interactive SHAP scenario-builder exposing the drivers behind each forecast across 7 features.

BCG Strategy Consulting: Telecom Company

- Conducted market analysis for handset leasing, projecting ~4% profitability growth, ~2% revenue increase and improved ARPU using method of comparables across US, Singapore, and Korea.

Mercedes Benz: In-Car Navigation Strategy

- Estimated a \$160-\$250M annual revenue opportunity by building a data-driven pricing model for proprietary in-car navigation.
- Mapped competitive landscape of premium Infotainment / Navigation services via Porter's forces and bundling growth lever for 33% profitability improvement.

Sony (Academic): Product Launch & Pricing

- Informed a B2B2C product launch business model by building a consumer and market trend analysis model from 3 years of fashion retail data.
- Projected €1.6B in 5-year revenue by designing a value-based subscription pricing strategy.

PWC Power BI: Telecom Company

- Informed customer retention HR strategies by designing Power BI dashboards for churn and diversity KPIs by analyzing 5K+ customer survey responses.

Network Analysis & NLP (Personal): Witcher Books Project

- Visualized an interactive Pyvis network and computed Louvain communities to show character's relational web after extracting entities from 1000+ lists in Witcher books using Python (Selenium, spaCy NER and Pandas).

Education

Master of Science (M.Sc.) – Management and Engineering CAME

RWTH Aachen University

CGPA: 1.7

Relevant Courses: Strategy Management, AI & Data Analytics, Finance Management, Marketing, Management & Engineering Perspectives

Oct 2022 – Mar 2026

Aachen, Germany

Bachelor of Technology (B.Tech.) – Mechanical Engineering

KJ Somaiya College of Engineering (Mumbai University)

CGPA: 1.4 (German grade equivalent)

Bachelor's Thesis: Amphibious Garbage Collector Robot (Designing and Simulation)

Aug 2016 – Oct 2020

Mumbai, India

Technical Skills

- **Analysis & Visualization:** Excel, Power BI, Tableau, Think Cell
- **Programming:** Python (Pandas, Numpy, Matplotlib), R (Tidyverse, base R), MySQL, MATLAB
- **AI Tools:** Claude, Gemini, Perplexity, ChatGPT (prompting, research synthesis, workflow speed)
- **Project Management & CRM:** SAP, S/4HANA, Confluence, Jira, MiroBoard, Agile/Scrum
- **Productivity & Technical:** PowerPoint, SharePoint, LaTeX, SolidWorks, AutoCAD

Languages

English (C1 Fluent) | Deutsch (B1 Intermediate) | Marathi (Native) | Hindi (Native)

Courses and Certificates

BCG	Strategy Consulting Job Simulation
PWC	Power BI Analysis
Fraunhofer LMS	E-Learning Scrum E-Learning Kanban
Google	Google AI Essentials Prompt Engineering
IBM	Python for Data Science Databases & SQL for Data Science
Dassault Systems	SolidWorks Associate

Hobbies

Film Critique | Team Sports (Football) | Drawing/Painting | Singing | Volunteering Cook at Tivoli Aachen